Course Project

In this semester-long course project, a team of students will work together, identify suitable healthcare data, analyze and visualize the data, and develop a machine learning prediction model or a generative AI tool. Teams are formed at the beginning of the semester per instructor's direction. The course project culminates in a team presentation and a final report.

Audience and Evaluation:

Imagine the audience for your report and presentation to be a company executive, government official, or other decision maker who will seek to take action based on the information you present. The action could be, for example, to go ahead with the implementation of your proposed AI solution, which may be a clinical or operative decision support tool using ML or Gen AI. You do not need to provide cost/budget estimations of your proposed solution, unless it is important to your specific project.

Dr. Wernz, as the course instructor, will be the one reading and evaluating your report, and will put himself in the shoes of the target audience. Dr. Wernz will also evaluate the appropriate application of technical tools and methods, and other important report and presentation elements as discussed in class.

Report:

Your report is limited to 6 pages.

Your first page should contain

- Project title
- Team number
- Names of students
- Target audience (what is the organizational role of the fictitious or real person for whom you are writing this report)
- Executive summary (apply Bottom Line Up Front - more on that during the course)

The report's first chapter/section begins on page 2, which means you have 5 pages of text and images beyond the cover page with the executive summary. References to be listed at the end, as part of the 6-page limit, following a consistent and conventional citation style. All pages must have page numbers.

All figures, graphs, and tables must be part of the 6 pages. You may reference external material, e.g., links to your interactive Tableau dashboard of which you show a screenshot in the report, your data sources, additional analysis results, and other relevant information, but do not rely on the reader to review them, so make sure that all important information you want to be evaluated on is contained in the 6 pages. Information of work referenced and described beyond the 6 pages, while maybe necessary for completeness or documentation references, will not improve your grade; but it can lower your grade if incorrect. External references allow the reader to confirm the quality and integrity of work and information provided.

The report and the presentation files are due at the same time, right before the presentations via Canvas submission (one submission per team).

Presentation:
Your presentations will occur during the final live class. Each team should present for 7-8 min on the assigned day. Each team member should be presenting. The audience of your presentation is the same as for the report:

Primary: The decision maker who is looking at your work to decide which action to take.
Secondary: Dr. Wernz, who will evaluate your technical approach and the team following course guidelines and key concepts.

You will receive detailed instructions and training on the recommended presentation and slide style.

Evaluation Criteria:

1. **Project Objective and Scope:**
   The project articulates a clear, well-defined, and meaningful objective, with an appropriate and focused scope. The problem is relevant, well-motivated, and demonstrates potential for real-world or practical impact.
2. **Technical Approach:**
   The project applies appropriate and technically sound data science methods aligned with the data and the stated objectives. This includes effective data wrangling and preparation, suitable use of visualization, and correct application of machine learning or gen AI, with justification for methodological choices.
3. **Presentation and Communication:**
   The project is communicated effectively through the presentation, slide deck, and written report, and adheres to all instructions regarding format, structure, page limits, and presentation time. The narrative is clear, logical, concise, and actionable. Slides follow best practices discussed in class, including strong action titles, Bottom Line Up Front structure, and coherent horizontal and vertical logic.

Project development milestones:

Date	Module #	Module Title	Week	Tasks completed
26-Aug			1	
02-Sep	1	**Health System and Health Data Fundamentals**	2	Team formation, schedule meeting for week 3
09-Sep			3	Team building and team charter: meet with your team, set weekly meeting schedule, discuss responsibilities, file sharing, etc.
16-Sep			4	Explore possible data source: week 3 live class provides overview and options
23-Sep	2	**Health Data Analysis**	5	Select your data set, and perform exploratory data analysis (EDA)
30-Sep			6	Analyze your data using Tableau (**individual assignment - data must be identified**)
07-Oct	3	**Health AI Models, Applications, and Implementation**	7	Finalize your team's data with refinements and additional data (if applicable)
14-Oct			8	Finalize your analysis/prediction objective, and write data section
21-Oct			9	Develop first AI/ML model
28-Oct			10	Refine/tune AI/ML model
04-Nov			11	If applicable: test and compare other models
11-Nov			12	Finalize AI/ML model, start writing report
18-Nov	4	**Health Careers and Success Factors**	13	Finalize report and slide presentation
02-Dec			14	Final Project Presentations